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Three Types of Tropical Waves Related to Tropical Cyclogenesis over the Western North Pacific

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Abstract The present study applies a space-time filter to identify three dominant types of tropical waves: Madden-Julian oscillations (MJOs), equatorial Rossby (ER) waves, and tropical depression (TD)-type disturbances. The impacts of these waves on tropical cyclones (TCs) were investigated based on 131 observations during the period 2000–07. The results suggest that 72% of TC geneses were related to the joint impacts of more than one type of wave. The composites for cases in different categories reveal that TCs related to the concurrence of the three types of waves have strong and large initial vortices at the time of TC genesis. In the absence of the MJO, ER- and TD-related TC genesis, embedded in easterly flow, exhibits a relatively fast initiation process and gives rise to a relatively small scale vortex. In contrast, without the ER wave contribution, TCs associated with ER and TD waves did not require strong convection at the time of genesis because an initial vortex can rapidly develop in the MJO active phase through persistent energy transfer. The MJO-related TC geneses were scattered in geographic distribution, as opposed to the clustered and eastward shift observed for genesis cases without contributions from MJOs.

Keywords: large-scale pattern, tropical waves, tropical cyclogenesis, western North Pacific

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1 Introduction

There is growing evidence that the formation of most tropical cyclones (TCs) over the western North Pacific (WNP) is significantly modulated by three types of tropical waves: Madden-Julian oscillations (MJOs), equatorial Rossby (ER) waves, and tropical depression (TD)-type disturbances. The MJO active phase provides large-scale cyclonic shear and convection due to lower-level westerly anomalies to create a favorable environment for tropical cyclone formation (Liebmann et al., 1994; Maloney and Hartmann, 2000, 2001; Hall et al., 2001; Kim et al., 2008). The 10–30-day westward-propagating intraseasonal oscillation in the tropical western Pacific, which is identified as an n=1 ER wave, can also influence cyclogenesis primarily by altering the mean flow in which the synoptic-scale and higher-frequency waves propagate (Molinari et al., 2007; Schreck and Molinari, 2009; Ko and Hsu, 2009). Off-equatorial synoptic disturbances with a 3–10-day period and a wavelength of roughly 3000 km, generally referred to as TD-type disturbances, can produce alternating cyclonic and anticyclonic circulation and trigger cyclogenesis under favorable conditions (Liebmann and Hendon, 1990; Takayabu and Nitta, 1993; Chang et al., 1996; Sobel and Bretherton, 1999; Li and Fu, 2006; Chen and Huang, 2009).

Previous studies have mainly considered the individual influences of these types of waves on tropical cyclogenesis over the WNP. However, to a large degree, two or more types of waves can coexist and jointly impact TC formation. The extent to which the three types of waves collaborate to form TCs and the variability in the circulation and convection related to these types of waves during the process of TC genesis are not well known. Studies on linkages of TC genesis with these three types of waves can help to improve our knowledge of the tropical atmosphere conditions for TC formation and may enhance the predictability of TCs on synoptic and intraseasonal scales.

2 Data and methodology

The data for TCs over the WNP spanning the period from 2000–07 was obtained from the Joint Typhoon Warning Center (JTWC). This study only considers the peak TC season from July to October, during which about 80% of annual total TCs are observed. In addition, signal contamination due to the seasonality of tropical waves can be diminished for this period. Daily variables from the National Centers for Environmental Prediction-National Center for Atmospheric Research (NCEP-NCAR) (Kalnay et al., 1996) were used for the analyses of large-scale circulation. They provide a horizontal resolution of 2.5°×2.5° on 17 pressure levels. Daily outgoing longwave radiation (OLR) data from the National Oceanic and Atmospheric Administration (NOAA) satellites on a 2.5° latitude-longitude grid were used as a proxy for deep tropical convection (Liebmann and Smith, 1996).

The space-time wave filter of Wheeler and Kiladis (1999) can extract tropical wave signals by filtering the wave period and truncating the wavenumber according to characteristics of wave propagation. A positive (or negative) wavenumber denotes eastward propagation (or westward propagation). The signals of three types of tropical waves are extracted as follows: a wavenumber ranging from 0 to 5 and a period ranging from 30 to 60 days for MJOs; a wavenumber from −1 to −10 and a pe-

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riod from 10 to 30 days for ER waves; a wavenumber from −6 to −14 and a period from 3 to 10 days for TD-type disturbances. These domains are in good agreement with the corresponding wavenumber-frequency of the three tropical waves in the previous studies (Roundy and Frank, 2004; Frank and Roundy, 2006). Based on distinct wave properties, the space-time filter can classify the different tropical waves into specific bins without overlap.

3 Results

To classify different categories, the time series of filtered OLR for MJO, ER, and TD wave bands were normalized, including 984 days over eight years (123 days each year). A wave type was considered active and contributed to one TC genesis if its normalized value was less than a threshold value at the time of the TC genesis (the first warning time in the TC dataset). The threshold value is defined as one fourth of the standard deviation, which is a conservative estimate, but representative of an active wave band. If the normalized OLRs for the three types of waves were simultaneously below individual thresholds at the time of TC genesis, the genesis in this case was attributed to MJO-ER-TD wave bands (assigned MET category). Similarly, if the TD wave was inactive and the other two waves were active, this case was classified into the MJO-ER (ME) category. The MJO-TD (MT) and ER-TD (ET) categories were determined similarly.

There were a total 131 TCs from July to October for the eight years. Among them, 30 genesis cases were classified into the MET category. This means that 23 percent of TC geneses were attributed to the joint impacts of the three types of wave. In contrast, there were 25, 19, and 20 cases in the ME, MT, and ET categories, respectively. Among the 131 cases, only one genesis case occurred when all three wave types were inactive, exhibiting a normalized OLR greater than the threshold value. All other TC geneses were merely attributed to one type of wave. Approximately 72 percent of TC geneses were related to the joint impacts of more than one type of wave.

Figure 1 shows the composite curves for the cases corresponding to individual categories, which depict the temporal evolution of unfiltered and filtered OLR before and after the TC genesis with respect to the location of genesis. The comparisons of different categories can help to qualitatively describe the individual roles of the three types of waves in the process of TC genesis. Figures 1a

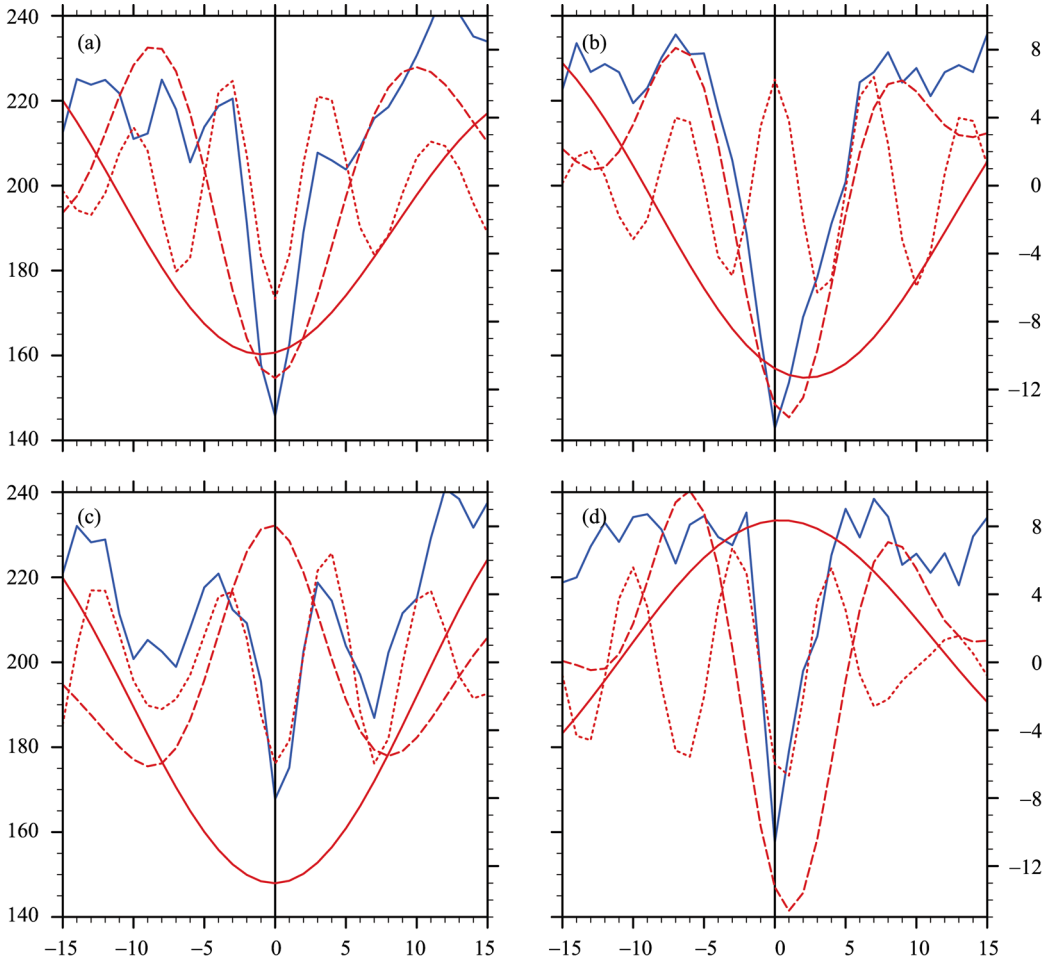


Figure 1 The time series of composited unfiltered (blue) and filtered (red) OLR in the TC formation location for the (a) MET, (b) ME, (c) MT, and (d) ET categories, with scales shown on the right and left, respectively. The red solid, dashed, and dotted line represent the MJO, ER, and TD wave bands in each plot. The abscissa denotes the lagging and leading day with respect to TC genesis time. The units are W m^{-2}.

and 1b reveal that unfiltered (total) OLR values were below 150 W m^{-2} during TC genesis in the MET and ME categories. In comparison, the initial convection during TC genesis in the other two categories was relatively weak, especially in the MT category (Fig. 1c) where the composite OLR reached 170 W m^{-2} at the time of TC genesis. This indicates that the TD disturbance during the convectively active phase of MJO readily developed from tropical cloud cluster to tropical depression without a further enhancement of convection, compared to the cases in the other three categories. In contrast, if the TD type did not contribute to TC formation (Fig. 1b), enhanced convection corresponding to MJO and ER wave bands were required, as indicated by the lowest OLR value among all the categories of 144 W m^{-2} during TC genesis. Although the averaged amplitude of the TD wave was the smallest (about −6 W m^{-2}), approximately half of the value of MJO and ER waves during TC genesis, the TD wave played a decisive precursor role in TC genesis by seeding the initial disturbance.

For the ET category, which lacks a favorable background related to MJO as shown in Fig. 1d, it was observed that TC genesis was rapid, as suggested by the sharp drop in unfiltered OLR from 230 to 160 W m^{-2} within two days. Otherwise, TC formation may have been inhibited by an unfavorable environment associated with an inactive MJO phase. Furthermore, a strong anomalous convection related to the ER wave type with a drop in OLR of about 15 W m^{-2} was required as a substitute for MJO to provide a favorable environment for the development of the TD disturbance. In addition, except for the cases in the MT category (Fig. 1c), ER wave anomalies were larger than the other two types of waves, suggesting that the ER type of wave acted as a primary contributor to total OLR, but required a relatively large amplitude to favor TC genesis.

The 850-hPa composite circulation and convection help shed light on the large scale environment prior to TC genesis. Figures 2–4 display composites of unfiltered wind and OLR to depict the circulation features from three days prior to cyclogenesis to the genesis. For the cases in the MET category, the westerly flow prevailed south of genesis location throughout the process of TC genesis, forming a horizontal cyclonic shear-like monsoon trough. Three days prior to cyclogenesis (Fig. 2a), two cyclonic centers collaborated with enhanced convection were anchored to the east and west of genesis location, implying the coexistence of eastward propagating MJO and westward propagating ER and TD waves. One day later, the scattered convection cells began to cluster and gave rise to an organized cyclonic system (Fig. 2b). The convection was further strengthened, forming a closed cyclonic circulation one day before genesis (Fig. 2c). At the time of tropical cyclogenesis, the cyclonic circulation was tightly contracted in scale and elliptically encircled the genesis location (Fig. 2d).

For the cases in the ME category, the circulation evolution resembled that of the MET category (not shown), except that enhanced convection related to the ER wave emerged on the eastern side of formation location before

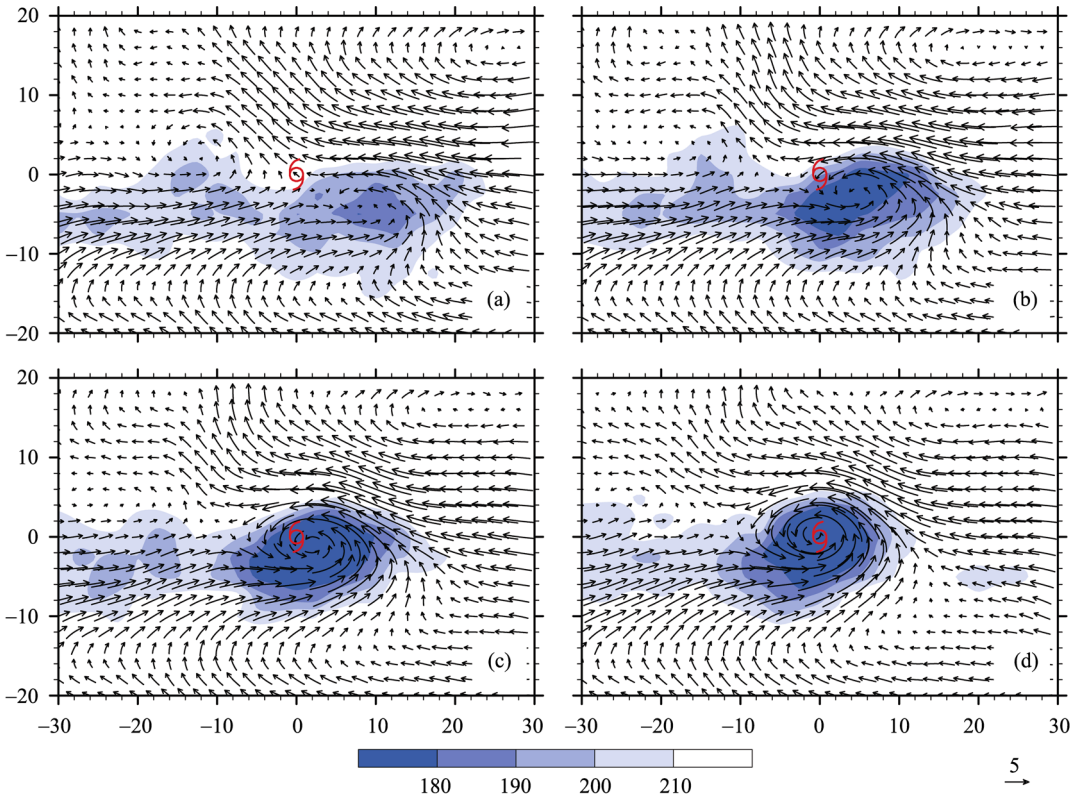


Figure 2 The composite unfiltered 850-hPa winds (vector, units: m s^{-1}) and OLR (shaded, units: W m^{-2}) for the MET category at (a) day 3, (b) day 2, (c) day 1, and (d) day 0 prior to TC genesis. The TC genesis locations are shown at the center of each panel.

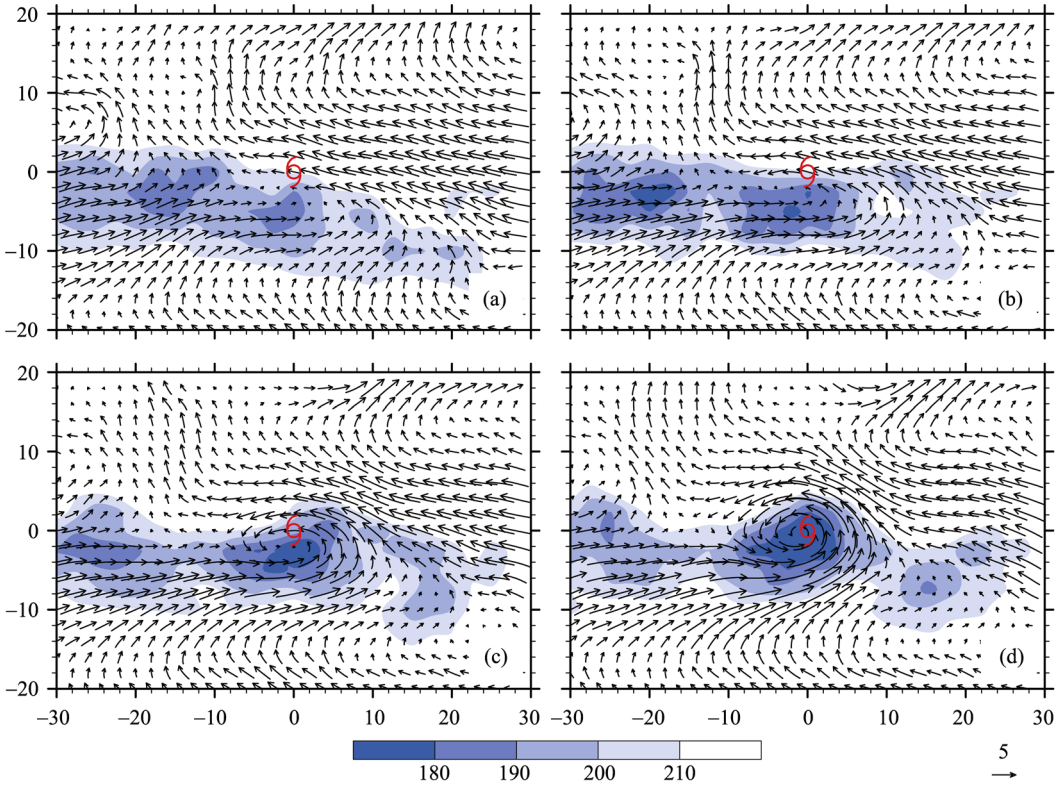


Figure 3 Same as Fig. 2, but for the MT category.

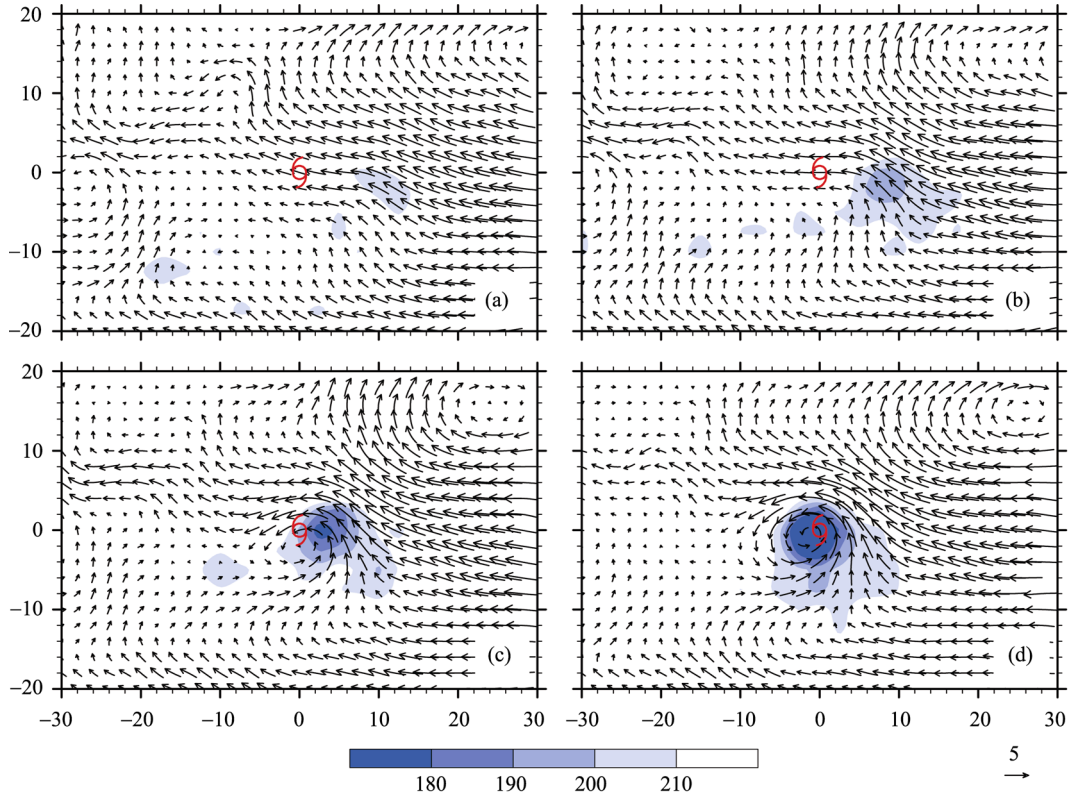


Figure 4 Same as Fig. 2, but for the ET category.

genesis due to a lack of contribution from the synoptic-scale TD wave type. These characteristics were also in accordance with the behaviors shown in Fig. 1b. In comparison with the above two categories, the elongated convection in the MT category with a northwest-southeast orientation spanned more than 50° longitude. The weak

TD signal began near the equator in the central Pacific three days before cyclogenesis (Fig. 3a). This suggests that synoptic disturbances that became TCs could be tracked to regions near the equator and date line, and may have origins from equatorial waves, which is in agreement with the previous studies (Lau and Lau, 1990; Dickinson and Molinari, 2002). Generally, TD disturbances propagated northwestward, exhibited an increase in amplitude through wave energy accumulation and contracted in spatial scale when they encountered the zonal wind convergence and cyclonic shear associated with large-scale MJO convection (Figs. 3b and 3c), eventually triggering TC genesis (Fig. 3d). This result is consistent with previous studies (Webster and Chang, 1988; Kuo et al., 2001).

The circulation evolution in the ET category displays distinct features (Fig. 4). Because of the absence of a favorable large-scale background related to MJO westerly phase, westward-propagating ER and TD waves were embedded in easterly wind and had a rather weak signal of convection to the east three days prior to genesis (Fig. 4a). With the OLR signal approaching the formation location, the convection rapidly intensified, but did not significantly change in the horizontal scale (Figs. 4b and 4c). At the time of genesis, a cyclonic system accompanied by compact convection, smaller in scale compared to the other three counterparts, was anchored in the formation location. The initial size and intensity during TC genesis, to a large degree, depended on the scale of tropical waves. Due to the active MJO phase with a large scale circulation and a long period, ER and TD waves embedded in convective envelope related to MJO can readily obtain energy from the MJO background, and trigger tropical cyclogenesis. When this occurred, the initial disturbance could grow into a large vortex through persistent energy transfer from the basic state. In comparison, ER and TD waves were characterized by small spatial scales and short temporal scales relative to MJO. As a result, the development of disturbance related to the ER and TD waves was confined to a limited spatial extent. Otherwise, the easterly and southeasterly winds poleward of genesis location were stronger than westerly and southwesterly winds equatorward in the ET category, which differed from the cases in the other categories of MJO contribution (Fig. 4d). Moreover, TCs in the ET category all formed east of 135°E and showed a clustered pattern compared to the scattered distribution for TC geneses in the other categories (not shown).

4 Summary

Previous studies have mainly focused on the linkage of tropical cyclone formation with individual types of tropical waves from the point of view of climatological statistics. In fact, a large fraction of TC formations are significantly modulated by large-scale patterns associated with the joint impacts of two or more types of tropical waves over the WNP. Using an effective space-time filter, the three dominant tropical waves (MJO, ER wave, and TD type disturbance) that are major contributors to TC genesis were extracted according to their wavenumber-frequency domain. This study investigated the combined influences of different types of waves and their related large-scale circulations on TC genesis by classifying TCs into four categories.

Among the 131 selected TCs, 72 percent of the TC generated were attributed to the collaborative contributions of more than one type of wave. For the cases in which all three wave types contributed jointly, the convection signals in the eastern and western flanks of TC formation location gradually approached each other and eventually lead to a strong and large TC vortex. In contrast, in the absence of an MJO contribution, TCs associated with ER and TD wave bands were embedded in an easterly flow and exhibited a rapid formation process. As a result, their initial vortices evolved from westward-propagating anomalous convections and were small on a spatial scale. Moreover, locations of TC genesis shifted eastward and were relatively clustered, as opposed to scattered distributions for the counterparts in the other categories. In the MT category without an active ER wave band, a TD disturbance could readily evolve into a TC under favorable conditions associated with the MJO active phase. Consequently, at the time of genesis, the TCs in the MT category did not exhibit strong convection compared to the other category cases.

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